Element B

Documentation/Analysis of Competitive Products

There are two ends of the cost spectrum—the cheapest[1] are around 11 dollars, while the most expensive[2] is around 600 dollars (based on Amazon). Many more are way more expensive, thousands of dollars, but are used only in specific hospital scenarios.

One example includes built up/large softer grip handles on utensils and tools to aid those who struggle with arthritis in their hands, to grip objects with ease and less pain. It was successful in reducing the amount of force required to grip those objects, by distributing the pressure of holding the object or tool over the large area of your hand. It helps lessen fatigue, aid in comfort and usability in repetitive tasks such as eating or cooking.[3] On the other hand, this will not actively ease the tension and pain associated with arthritis, and will only work on certain objects that can have a modified handle.

For instance, these Imak gloves[4] offer mild compression designed to improve circulation, reduce swelling, and ease pain in the hands and fingers. Although it is very affordable, many users report insufficient long-term pain relief, as the compression alone doesn't address severe stiffness or dexterity issues. The glove is also not very durable. They are fingerless gloves, made out of soft, breathable cotton. It is successful in reducing swelling in the hands, warming up the joints, and improving circulation in the hands. Because it is also fingerless, the glove makes it easier to still continue with daily activities while wearing the gloves. It was not successful at actively helping and assisting patients with arthritis with their grip.

Thermoskin gloves[5] use a thermal material that retains natural body heat, aiming to relieve stiffness and increase mobility in arthritic hands without an external energy source. While it is good at providing soothing warmth and mild compression, it can cause sweating and discomfort with prolonged wear; not breathable. They do not provide active mechanical support for grip strength or fine motor assistance. The glove was not successful at providing comfort for long periods of time, due to the inability for excess sweat to evaporate out of the glove, and the lack of breathability due to the fabric choice. The glove was successful in retaining body heat that could be used to lessen stiffness in the joints.

Smart/Powered assistive gloves, like Neofect Smart Glove and SensoGlove prototypes, are designed for rehabilitation or/and grip assistance using sensors, robotics, or soft actuators. They track hand movement and provide resistance or assistance for therapy. These gloves do offer ways to actively support movement and measure progress digitally–like through app connectivity. However, this glove is on the expensive side ($300–$1,000+), bulky, and not practical for daily wear. These types of gloves are targeted more as tools used at rehabilitation centers rather than everyday use. This design was unsuccessful at designing a product to actively assist people with arthritis on a daily basis and therefore an affordable product for daily life. It was also unsuccessful at massaging and heating the joints of the hand to provide comfort. However, the design was successful at assisting the grip of those with arthritis to help with therapy and in measuring progress digitally, giving patients and their caretakers an idea of how successful the therapy is.

Dr. Arthritis Copper Gloves are an example of an existing solution to pain and discomfort in the hands due to arthritis. These arthritis copper gloves are compression gloves that have copper threaded with the fabric of the glove in order to decrease inflammation via natural properties and to help with joint mobility in the hands. This solution was successful in making the gloves lightweight, affordable, and fit like a “glove”. However, this product is unsuccessful as the evidence behind copper’s support in being anti-inflammatory is weak. It was successful in using fabric to assist in passive compression, however the compression was mediocre. It was successful at helping joint pain, albeit not sufficiently. Users didn’t report any significant improvement in pain management after wearing the gloves (Mayo Clinic, 2021; Arthritis Foundation, 2023).

Works Cited

1. “Medically Developed Arthritis Compression Handbook.” Amazon.nl, https://www.amazon.nl/Medically-Developed-Arthritis-Compression-Handbook/dp/B06XW3GS9X/.
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5. “Infrared Arthritis Gloves – Half Finger.” Veturo Therapy, https://www.veturotherapy.com/shop-infrared-therapy/infrared-arthritis-gloves-half-finger/.